DOD-P-21035A(NAVY)
21 November 1977
SUPERSEDING
MIL-P-21035(SHIPS)
23 August 1957
(See 6.6)

METRIC

MILITARY SPECIFICATION

PAINT, HIGH ZINC DUST CONTENT, GALVANIZING REPAIR (METRIC)

This specification is approved for use by all interested Commands of the Department of the Navy and the Marine Corps and is available for use by all other Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a high zinc dust content paint for regalvanizing welds in galvanized steels.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

RR-S-366 - Sieve, Test.
TT-P-460 - Pigment, Zinc-Dust (Metallic-Zinc-Powder) Dry.
PPP-P-1892 - Paint, Varnish, Lacquer, and Related Materials; Packaging, Packing, and Marking of.

MILITARY

MIL-S-16113 - Steel Plate, High Tensile(HT), Hull and Structural.

STANDARDS

FEDERAL

FED-STD-141 - Paint, Varnish, Lacquer, and Related Materials; Methods of Inspection, Sampling, and Testing.
FED-STD-313 - Material Safety Data Sheets, Preparation and the Submission of.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 93 - Flash Point by Pensky-Martens Closed Tester, Test for.
D 521 - Zinc Dust (Metallic Zinc Powder), Chemical Analysis of.
D 562 - Consistency of Paints Using the Stormer Viscosimeter, Test for.
D 1475 - Density of Paint, Varnish, Lacquer, and Related Products, Test for.
D 2369 - Volatile Content of Paints, Test for.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Ship Engineering Center, SEC 6124, Department of the Navy, Washington, DC 20362 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadephia, Pennsylvania 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Toxicity. The material shall have no adverse effect on the health of personnel when used for its intended purpose (see 4.6). Questions pertinent to this effect shall be referred by the procuring activity to the appropriate service medical department which will act as advisor to the procuring activity.

3.2 Materials. The paint shall consist of zinc dust conforming to type I of TT-P-460, either ready-mixed or in a two-compartment container with resins, oils, and solvents properly processed to conform with all requirements as herein specified.

3.3 Manufacture. When the paint is supplied as a ready-mixed, single-package product, the component raw materials shall be mixed and ground as required to produce a product which is uniform, free from dirt and grit, entirely suitable for the purpose intended, and in full conformity with the requirements of this specification.

3.4 Quantitative requirements. Quantitative requirements shall be as shown in table I and as herein specified. Test reports required by method 1031 of FED-STD-141 shall include the exact formula used.

TABLE I. Quantitative requirements.

Characteristic	Minimum	Maximum
Pigment, percent of weight of nonvolatile content	94.0	---
Pigment, percent zinc by analysis	97.5	---
Pigment, kg (pounds) per gallon of paint	5.455 (12.0)	---
Flash point, °C(°F)	38 (100)	---
Drying time - set-to-touch, hours dry hard, hours	0.5 ---	2 8

3.4.1 Solvent. The solvent portion of the paint shall conform to requirements herein specified.

(a) A combination of hydrocarbons, alcohols, aldehydes, ethers, esters, or ketones having an olefinic or cycloolefinic type of unsaturation except perchloroethylene: 5 percent maximum.
(b) A combination of aromatic compounds with eight or more carbon atoms to the molecule except ethylbenzene, methyl benzoate, and phenyl acetate: 8 percent maximum.
(c) A combination of ethylbenzene, ketones having branched hydrocarbon structures, trichloroethylene, or toluene: 20 percent maximum.

3.5 Qualitative requirements. The paint shall comply with the qualitative requirements specified herein.

3.5.1 Condition in container. The mixed paint or the vehicle portion, if supplied as a two-compartment package, shall be capable of being broken up to a smooth, uniform consistency. It shall not increase more than 15 Krebs units in viscosity or increase more than 2 hours in drying time. It shall not liver, curdle, gel, or show any other objectionable properties for at least 1 year after date of manufacture, when tested as specified (see 4.4).

3.5.2 Application characteristics. The mixed paint, when tested as specified (see 4.4.4), shall be capable of being brushed out without excess drag on the brush. When dry, the brush-coated surface shall be free from sags, runs, wrinkles, excess brush marks, or other film defects. The film shall exhibit good adhesion and a smooth, uniform appearance.

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3.5.3 Salt spray resistance. The uncoated area of the test panel shall not be completely rusted. The coated area shall show less than 5 percent rust. There shall be no blistering, loss of adhesion, or other film failure, except pinpoint rusting and white corrosion product, when tested as specified (see 4.4.3).

3.5.4 Knife test. A film of the mixed coating, when prepared and tested as specified (see 4.4.5), shall be tough and hard and shall adhere tightly to the metal panel. It shall be difficult to furrow off with the knife and shall not flake, chip, or powder. The knife cut shall show bevelled edges.

3.5.5 Stability in partially full container. When tested as specified (see 4.4.6), the vehicle reaction shall exhibit no evidence of skinning.

3.5.6 Material Safety Data Sheet. The procuring activity shall be provided a material safety data sheet (MSDS) at the time of contract award. The MSDS is DD form 1813 and found in and part of FED-STD-313. The MSDS shall be included with each shipment of the material covered by this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection. Quality conformance inspection shall be provided in accordance with method 1031 of FED-STD-141 and as herein supplemented.

4.3 Additional inspection.

4.3.1 Ingredient materials. When requested by the testing laboratory or other controlling authority, 1 pint of each ingredient used (see 3.4) shall be supplied for test purposes.

4.4 Test procedures. Tests shall be conducted in accordance with table II.

TABLE II. Test procedures.

Tests[1]	Applicable method of FED-STD-141	Applicable ASTM test method
Pigment content[2]	4021	----
Nonvolatile content[2]	----	D 2369
Application characteristics	2141	----
Condition in container	3011	----
Drying time	4061	----
Weight per gallon[2]	----	D 1475
Viscosity, Krebs-Stormer	----	D 562
Flash point[2]	----	D 93
Salt spray resistance	6061	----
Knife test	6304	----
Total zinc, percent of pigment[2]	----	D 521
Stability in partially full container	3021	----

1/ These tests shall be performed and reported even though they may not be specific requirements of this specification.

2/ For ready-mixed paint, determine percent pigment, nonvolatile content, percent zinc, weight per gallon, and flash point. Calculate requirements of table I. For two-compartment packaged paint, the nonvolatile content and flash point on the vehicle portion and total zinc on the pigment portion may be determined without mixing the components, provided the pigment portion consists solely of dry pigment. The mixing instruction may be used in calculating the requirements of table I.

4.4.1 Pigment analysis.

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4.4.1.1 Pigment content. Extract the pigment from a weighed sample of the mixed paint as in method 4021 of FED-STD-141, using the appropriate extraction mixture. Dry and weigh the extracted pigment. Calculate the percent pigment in the paint.

4.4.1.2 Total zinc. Determine the total metallic zinc content in the extracted pigment or in the zinc dust pigment of the two-component product in accordance with ASTM D 521.

4.4.2 Drying time. Determine drying time by method 4061 of FED-STD-141, except that the specified conditions of temperature and humidity shall apply only for referee tests in case of dispute. All other tests shall be conducted under prevailing laboratory conditions.

4.4.3 Salt spray resistance. Determine salt spray resistance in accordance with method 6061 of FED-STD-141, except that three 10.6-centimeter (-cm) by 15.24-cm by 0.32-cm (4-by 6-by 1/8-inches) mild steel panels shall be used for this test. All surfaces shall be blasted with a medium grit [pass through 0.071-cm (710-micron) and retained on 0.042-cm (420-micron) sieve conforming to [ ]-366] prior to coating. A 1.27-cm (1/2-inch) wide strip shall be left uncoated along the longitudinal center of test face of each panel to observe degree of cathodic protection provided by the coating. Two coats of paint shall be brush applied uniformly to all remaining surfaces of each panel to a minimum thickness of 5.08 millimeter (mm) (2 mils). A drying time of 24 hours between coats and 48 hours for the second coat shall be observed. The panels shall be exposed in the salt spray apparatus for 288 hours, then evaluated for percent rust on the coated and uncoated areas. Rust streaking shall be disregarded in determining these percentages. The average of the three panels shall be determined for each area. Additionally, film failure of coated areas shall be noted and reported.

4.4.4 Application characteristics. Determine application characteristics of the mixed paint in accordance with method 2141 of FED-STD-141. The paint shall be applied to 30.48-cm (1-foot) squares of 0.64-cm (1/4-inch) galvanized steel plates conforming to MIL-S-16113. The film thickness of the dried paint shall be averaged from five measurements using a micrometer or other equivalent instrument. Check for compliance to 3.5.2.

4.4.5 Knife test. Solvent clean a steel panel of the type specified in method 2011 of FED-STD-141, using the petroleum naphtha-ethylene glycol monoethyl ether mixture. Draw down a 5.08-cm (2-inch) wide film of the mixed paint with a suitable film applicator designed to deposit a dry-film thickness of 0.00229 cm to 0.00279 cm (0.0009 to 0.0011 inch). Allow the paint film to dry for 48 hours in a horizontal position at ambient laboratory conditions. Perform the knife test in accordance with method 6304 of FED-STD-141 and observe for compliance with 3.5.4.

4.4.6 Stability in partially filled container. Check by method 3021 of FED-STD-141 for compliance to 3.5.5.

4.5 Inspection of preparation for delivery. Inspect the packaging, packing, and marking of the material to determine compliance with the requirements of section 5 of this specification.

4.6 Toxicity. A manufacturer of material shall disclose the formulation of his product to the Navy Bureau of Medicine and Surgery, Navy Department, Washington, DC 20372. The disclosure of proprietary information, which shall be held in confidence by the Bureau of Medicine and Surgery, shall include: the name, formula, and approximate percentage by weight and volume of each ingredient in the product; the results of any toxicological testing of the product; identification of its pyrolysis products; and any such other information as may be needed to permit an accurate appraisal of any toxicity problem associated with the handling, storage, application, use, or disposal of the material.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurement.)

5.1 Packaging, packing, and marking. The paint shall be packaged, packed, and marked in accordance with PPP-P-1892. The level of packaging shall be A or C and level of packing shall be A, B, or C, as specified (see 6.2). The product shall be furnished in 1-gallon cans or 5-gallon pails of the mixed paint or in a two-compartment kit containing separately predetermined quantities of zinc dust and vehicle, as specified (see 6.2).

5.1.1 Special marking. In addition to the markings required by the contract or order (see 6.2), each container, interior and exterior, shall be marked with the following:

"The volatile content of the material in this container is not photochemically reactive as defined by Rule 102 of the South Coast Air Quality Management District." (See 6.4)

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6. NOTES

6.1 Intended use. This specification covers paint intended for use for regalvanizing welds in galvanized steels.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type of product required (ready-mixed or two-component) (see 3.2).
(c) Level of packaging and level of packing required (see 5.1).
(d) Size of container required (see 5.1).
(e) Special marking required (see 5.1).

6.3 Unit of purchase. Paint should be purchased under this specification by volume, the unit being 1 U.S. gallon (231 cubic inches) at 15.5°C (60°F).

6.4 Packaging. Level B is intended to provide economical but limited protection. It should be specified only when it is determined that the paint will be held in covered storage no more than 1 year from date of initial packaging.

6.5 Volatile content. Although the container marking specifically refers to the South Coast Air Quality Management District, the paint may be used anywhere else a paint complying with 3.3.1 is allowed. This includes all other air pollution control districts or similar areas controlling the emission of solvents into the atmosphere. Information regarding Los Angles County Air Pollution Rules 102, 442, and 443 may be obtained from: South Coast Air Quality Management District, Metropolitan Zone, 434 South San Pedro Street, Los Angeles, California 90013.

6.6 Changes from previous issue. The symbol "#" is not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodian:
Navy - SH

User activity:
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Preparing activity
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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
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5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	d. DATE OF SUBMISSION *(YYMMDD)*

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